Hypovolemia

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Introduction

Hypovolemia is a generic term encompassing volume depletion and dehydration.[1] Volume depletion is the loss of sodium-containing fluid from the intravascular and interstitial spaces and is more frequently associated with hypotension and tachycardia than is dehydration. Dehydration, such as from excess sweating, implies loss of water from both extracellular (intravascular and interstitial) and intracellular spaces and leads to elevated plasma Na^+ and osmolality. Causes of hypovolemia include hemorrhage; volume losses from GI tract, kidneys, skin or respiratory tract; and fluid sequestration or third-space losses (loss of intravascular volume). The focus of this chapter is the treatment of adults. For treatment of children, see Dehydration in Children.

Goals of Therapy

- Restore volume to relieve symptoms and prevent organ damage

Investigations

- History:
  - symptoms of hypovolemia include thirst, fatigue, muscle weakness, palpitations and postural lightheadedness
  - external source of loss: bleeding, vomiting, diarrhea, excessive diuresis or perspiration; enquire about drugs (see Table 1) or other toxins that might be contributing to fluid loss
  - presentation in older adults can be nonspecific
- Physical exam:
  - determine the presence and severity of hypovolemia and rule out hemorrhage
  - pulse: heart rate (HR) increase of more than 30 beats/min after 1 minute of standing from recumbent position is the most accurate sign of moderate to severe volume depletion[1] (note that presence of a beta-blocker will limit the expected increase in heart rate caused by volume loss)
  - blood pressure: postural decline of systolic pressure of more than 20 mm Hg suggests volume depletion;[1] moderate to severe volume loss may lead to persistently low blood pressure, even while supine
  - dry axillae support the diagnosis of hypovolemia;[1] moist mucous membranes and axillae argue against hypovolemia; decreased skin turgor, prolonged capillary refill time and decreased eyeball tension are late signs (see Dehydration in Children) and are insensitive in adults[3]
- Laboratory tests:
  - blood: in hypovolemia, hematocrit and albumin concentrations increase and urea increases disproportionately to creatinine; sodium concentration may be normal, low or high and is dependent on the type and amount of fluid consumed by the patient in response to thirst
  - urine: hypovolemia is suggested if urine volume is <0.5 mL/kg/h, if Na^+ is <20 mmol/L or if osmolality is >450 mOsm/kg;[4] urine specific gravity (SG) is less accurate but SG >1.020 suggests concentrated urine and may indicate hypovolemia

***Table 1:** Drugs That May Alter Hydration Status*[2]

Drug or Drug Class	Mechanism
Anticholinergics, antipsychotics, anxiolytics	Altered central thermoregulation
Digoxin, fluoxetine, metformin	Decreased appetite
Trihexyphenidyl	Decreased sweat production
ACE inhibitors, benzodiazepines, SSRIs	Decreased thirst sensation
Antibiotics, colchicine, laxatives, lithium, PPIs	Diarrhea
Diuretics, SGLT2 inhibitors	Increased urine volume

Abbreviations: ACE = angiotensin converting enzyme; PPI = proton pump inhibitor; SGLT2 = sodium-glucose cotransporter 2; SSRI = selective serotonin reuptake inhibitor

Therapeutic Choices

Therapy is designed to restore volume while replacing ongoing losses and addressing the primary cause (see Figure 1). Concurrent fluid and electrolyte disorders will influence rate of replacement and choice of fluid used (see Table 2). The severity of hypovolemia is expressed as a percentage of total body weight, assuming each litre of fluid lost weighs 1 kg.

- Hypovolemia suspected but uncertain: Consider an IV fluid challenge of 250–500 mL of crystalloid over 30 minutes. Improvement in HR and BP is expected if low cardiac output is due to hypovolemia.
- Mild hypovolemia (<5% of body weight): Fluid loss is <10% of plasma volume and oral therapy is usually adequate. Water, juices, soft drinks or soup broth with extra salt or a commercially available electrolyte solution (see Dehydration in Children) may be used.[5] Rice-based oral solutions have proven effective for diarrheal conditions in developing countries.
- Moderate (5–10% of body weight) or severe (>10% of body weight) hypovolemia or inability to ingest oral fluids: IV therapy is required and 4 types of solutions are used:
  - **Isotonic crystalloid solutions** (sodium chloride 0.9% or Ringer lactate) are the first choice for initial treatment of acute hypotension and tachycardia associated with volume depletion. They redistribute to extracellular fluid. For every litre infused, up to 1/3 will stay in the intravascular space and the remainder goes to the interstitial space. Ringer lactate is considered a balanced crystalloid because it has a sodium, potassium and chloride content closer to that of intravascular fluid.
  - **Blood products** are effective intravascular expanders. They are indicated for active bleeding in hemorrhagic hypovolemia.
  - **Dextrose 5% in water** (D5W) distributes throughout total body water and is useful for intracellular volume loss (dehydration). It is a poor plasma volume expander, as very little remains in the intravascular space; therefore, it is not useful for managing acute hypotension or tachycardia.
  - **Colloid solutions**, including albumin, dextran, gelatins (not available in Canada) and starch compounds, are better intravascular volume expanders than crystalloid because they remain in the intravascular space longer than crystalloids. However, there is little evidence of benefit and their cost is high, so their use as the primary fluid is not justified in hypovolemic states.

Determining Fluid Requirements

There is no precise formula since disease, age, source and rate of fluid loss influence fluid requirements. The amount and composition of replacement fluid chosen will depend on the type of fluid loss and assessment of the patient's individual needs. Fluid prescribing should be given the same status as drug prescribing.[6]

In nonhemorrhagic hypovolemia with obvious hemodynamic compromise, begin with at least 1 L of crystalloid over 30 minutes and a second litre over the next hour. Closely monitor HR, BP, peripheral perfusion and jugular venous pressure, watching for improvement or fluid overload. Delay in volume repletion can lead to ischemic injury and multiorgan system failure. In less severe hypovolemia, give 250–500 mL/h of crystalloid. Continue fluid administration until treatment goals are achieved.

For burns, the Parkland formula (4 mL/kg/% total body surface area burned) can guide fluid resuscitation. Crystalloid solutions, like Ringer lactate, are the standard initial fluid for burn resuscitation. Plasma, particularly fresh frozen plasma, may be a beneficial adjunct or even a primary resuscitation fluid in certain cases due to its ability to address the endotheliopathy of burns and potentially reduce the overall fluid requirements.[7]

Isotonic crystalloids are recommended for fluid resuscitation of critically ill patients, including those with sepsis.[8][9] A study comparing 4% albumin (colloid) with normal saline (crystalloid) for fluid resuscitation in ICU patients yielded similar outcomes, including 28-day mortality;[10] colloid solutions are not recommended for acute resuscitation due to their higher cost and lack of superiority over crystalloids.[9] The latest evidence suggests there may be a benefit to using a balanced crystalloid solution such as Ringer lactate in resuscitation of critically ill and noncritically ill patients.[11][12] Due to safety concerns,[13] use of starch compounds should be avoided in critically ill patients, particularly those with sepsis, as they increase the risk of acute kidney injury, need for renal replacement therapy and mortality.[14][15]

Maintenance fluids must be administered in addition to those given to correct the deficit. In adults, maintenance is possible with approximately 30 mL/kg/day or 2000–2500 mL/day containing approximately 75 mmol of Na^+ and 50 mmol of K^+ in a 24-hour period. This could be provided, for example, by using dextrose 3.3% with sodium chloride 0.3% and 20 mmol K+ added to each litre.

For hemorrhagic hypovolemia, early use of blood products over crystalloid resuscitation results in better outcomes. Balanced transfusion using 1:1:1 or 1:1:2 ratio of plasma to platelets to packed red blood cells results in better hemostasis.[16] In the CRASH-2 trial,[17] administration of **tranexamic acid** to patients with severe bleed within 3 hours of traumatic injury appears to decrease death from a major bleed. A single intravenous dose of 2 g tranexamic acid can be infused over 20 minutes. Research on oxygen-carrying substitutes as an alternative to red blood cells is ongoing, and no blood substitutes have been approved for use in North America.[18]

Other commercially available crystalloid solutions are more costly and, aside from dextrose 3.3% with sodium chloride 0.3% (2/3–1/3) or dextrose 5% with sodium chloride 0.45% (D5-1/2 NS), they play little role in managing volume depletion. Users of other crystalloids must be aware of their contents. For example, Ringer lactate contains calcium, potassium and lactate in addition to sodium and may not be suitable for some patients, e.g., those with renal dysfunction.

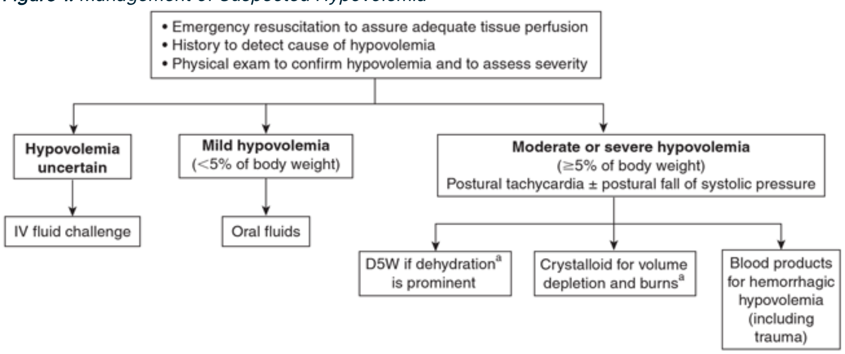
Therapeutic Tips

- If presence of hypovolemia is uncertain, consider a fluid challenge.
- Crystalloid is the fluid of first choice for patients with hypotension and tachycardia associated with volume depletion.
- Blood products are the first choice fluid for patients with hypotension and tachycardia associated with active hemorrhage.

- Dextrose in water is a poor plasma volume expander.
- Additional potassium may be required for fluid loss associated with diarrhea, vomiting or over-diuresis (see Potassium Disturbances).
- Colloid solutions are falling out of favour due to the adverse effects associated with their use.
- The toxicity of certain radiocontrast agents and drugs (e.g., ACE inhibitors, ARBs, metformin, sodium-glucose cotransporter 2 inhibitors, NSAIDs) is enhanced during hypovolemia and they should be withheld until volume is restored.

Algorithms

Figure 1: Management of Suspected Hypovolemia



[a] Dehydration refers to loss of intracellular water leading to elevated plasma sodium and osmolality. Patients with dehydration may or may not have hypotension and tachycardia. Volume depletion results from loss of salt and water from the extracellular space. Volume-depleted patients exhibit circulatory instability.

Abbreviations: D5W = dextrose 5% in water

Drug Table

Table 2: Intravenous Solutions for Hypovolemia

Drug/Cost	Indications	Adverse Effects	Comments
Drug Class: Blood Products			
frozen plasma	Major hemorrhage, burn or microvascular bleeding.	Fluid overload. Transfusion-related acute lung injury (TRALI) (rare).	One unit contains 400–900 mg fibrinogen.
platelets	Major hemorrhage, platelet dysfunction or marked bleeding.	Platelets are stored at 20–24°C, so there is a small risk of bacterial contamination.	One unit contains approximately 30×10^{10} platelets. Shelf-life is 5–7 days.
red blood cells (RBCs)	Hemorrhage.	Fluid overload. Rare: febrile nonhemolytic transfusion reaction (0.3% per unit of RBC).	Replaces or increases oxygen-carrying capacity.
Drug Class: Crystalloids			
dextrose 5% in water (D5W) generics	Dehydration.	Interstitial edema, hyperglycemia.	Poor plasma volume expander. Water is distributed to both intracellular fluid and extracellular fluid after dextrose metabolism.
Ringer lactate (Na^+ 130 mmol/L) Lactated Ringer Injection, generics	Initial treatment of hypovolemia.	Fluid overload, interstitial edema, dilutional coagulopathy, hyponatremia; may aggravate pre-existing hyperkalemia.	Contains Ca^{++} 1.4 mmol/L, K^+ 4 mmol/L, Cl^- 109 mmol/L, lactate 28 mmol/L. pH 6.4
sodium chloride 0.9% (normal saline or NS; Na^+ 154 mmol/L) generics	Initial treatment of hypovolemia.	Fluid overload, interstitial edema, dilutional coagulopathy. Hyperchloremic metabolic acidosis with infusion of large amounts of NS (usually >3 L). Hypernatremia.	Intravenous solutions with Na^+ concentrations that approximate normal serum Na^+ cause more intravascular and interstitial expansion than D5W solutions. pH 5.7

Drug/Cost	Indications	Adverse Effects	Comments
dextrose 5% with sodium chloride 0.9%, (D5W-NS; Na⁺ 154 mmol/L) generics	Hypovolemia and dehydration.	Fluid overload, interstitial edema, dilutional coagulopathy. Hyperchloremic metabolic acidosis with infusion of large amounts of NS (usually >3 L). Hyperglycemia.	Intravenous solutions with Na^+ concentrations that approximate normal serum Na^+ cause more intravascular and interstitial expansion than D5W solutions. Compared with NS or Ringer lactate, D5W results in comparatively small changes in plasma volume because it distributes throughout the total body water space, including the intracellular space.
sodium chloride 0.45% (half-normal saline or 1/2 NS; Na^+ 77 mmol/L) generics	Maintenance fluids.	Fluid overload, interstitial edema, dilutional coagulopathy. Hyperchloremic metabolic acidosis with infusion of large amounts of NS (usually >3 L).	Intravenous solutions with Na^+ concentrations that approximate normal serum Na^+ cause more intravascular and interstitial expansion than D5W solutions.
dextrose 3.3% with sodium chloride 0.3%, 2/3–1/3 (Na^+ 51 mmol/L) generics	Maintenance fluids.	Fluid overload, interstitial edema, dilutional coagulopathy. Hyperchloremic metabolic acidosis with infusion of large amounts of NS (usually >3 L).	Intravenous solutions with Na^+ concentrations that approximate normal serum Na^+ cause more intravascular and interstitial expansion than D5W solutions.
dextrose 5% with sodium chloride 0.45%, D5W-1/2NS (Na^+ 77 mmol/L) generics	Maintenance fluids.	Fluid overload, interstitial edema, dilutional coagulopathy. Hyperchloremic metabolic acidosis with infusion of large amounts of NS (usually >3 L). Hyperglycemia.	Intravenous solutions with Na^+ concentrations that approximate normal serum Na^+ cause more intravascular and interstitial expansion than D5W solutions.
Drug Class: Colloids			
albumin Alburex 5, Alburex 25, Plasbumin-5, Plasbumin-25, generics	Volume depletion. Contraindicated in patients at risk of circulatory overload, such as those with history of heart failure, renal insufficiency or stabilized chronic anemia.	Fluid overload, anaphylactoid/anaphylaxis (rare).	Albumin is a blood product. Volume remains within the extracellular compartment. Albumin 5% is iso-oncotic (does not pull fluid into the intravascular space) with normal plasma; intravenous infusion will expand circulating blood volume by an amount approximately equal to the amount infused; does not aggravate tissue dehydration; concomitant use of crystalloids may be necessary to maintain fluid balance. Albumin 25% will expand plasma volume 3–4 times the volume infused. There is little evidence to support the routine use of albumin in

Drug/Cost	Indications	Adverse Effects	Comments
			managing hypovolemia. [19]
dextran 40 in dextrose 5% or normal saline, respectively Dextran 40 in Dextrose Injection, Dextran 40 in Sodium Chloride Injection	Volume depletion.	Aggravation of bleeding; interference with blood coagulation; anaphylaxis with higher molecular weight solutions.	Not commonly used as plasma expander.
hydroxyethyl starch (tetrastarch) 6% in isotonic electrolyte or normal saline, respectively Volulyte, Voluven	Volume depletion.	Bleeding, fluid overload, liver failure, renal injury, pruritus; macroamylase formation may lead to incorrect diagnosis of pancreatitis because of possibility of increase in blood amylase; anaphylactoid reaction (rare).	Not commonly used as plasma expander. Contraindicated in patients with sepsis, severe liver disease or renal impairment (anuria and oliguria) not related to hypovolemia.

Suggested Readings

Callum JL, Pinkerton PH, Lin Y et al. (2023). *Bloody easy 5.1: blood transfusions, blood alternatives and transfusion reactions: a guide to transfusion medicine* [PDF file].

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